Portable Test Systems (Color)
PTSx.xC
RS232 Interface Description

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1. ⚠ Safety

The following symbol appears on the product and in the operation manual with the meaning:

⚠ **Caution! Please consult the operation manual before using the instrument.**
Failure to follow or carry out instructions preceded by this symbol may result in personal injury or damage of the device and the installation.

⚠ General precautions for use ⚠

To prevent electric shock:

- **This product has to be only used by qualified personnel practicing applicable safety precautions.**
- **Use caution during installation and use of this product; high voltages and currents may be present in circuit under test.**
- **Local safety regulations must be observed.**

2. General

The PTS color portable test systems consists of an integrated three-phase or one-phase current source and optional a one or three-phase voltage source. Also integrated is a one or three-phase electronic reference standard of accuracy class 0.05%. Characteristic features of the PTS color systems are its wide measuring range, high accuracy and high tolerance to unwanted external influences.
A PTS color system allows the monitoring of meter installations as well as analysis of the local mains conditions.

Advantages

- Easy verification of meters under precise load conditions, using the built-in, compact, current and voltage source
- Automatic operation with predefined load points without the need for an external PC
- Exchangeable Compact Flash (CF) memory card for measurement results and customer data
- Display of vector diagram and phase sequence for analysis of the supply conditions
- User-friendly system for data input and operation of combined source and reference meter
- The system may be used either as a stand-alone reference standard meter, or together with the integrated power source

Functions

- Independent generation of single or three-phase loading conditions for verification of meters
- Active, reactive and apparent energy measurement for three phase, different wire, systems with integrated error calculator and pulse output
- Vector diagram, harmonics spectrum, wave form and rotary field display for analysis of the mains conditions
- Burden measurement of Current Transformer (CT) and Potential Transformer (PT)

Application

- On site meter measurements
- Verification of energy registration
- Verification of the circuit load conditions

Options

- Software CALSOFT
- Error compensated clip-on CT's up to 100 A
- Error compensated clip-on CT's up to 1000 A
- Flex3000 up to 3000A

For any further information concerning the use of this instrument, or in the event of any questions, you are invited to contact EMH / MTE directly at:

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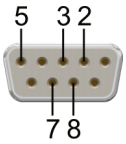
3. Connection and start-up

3.1 RS232 Connector

Serial data interface RS232
Connector: SUB-D socket 9-pole



Pin 2 ⇨ TxD
Pin 3 ⇨ RxD
Pin 5 ⇨ GND
Pin 7 ⇨ CTS
Pin 8 ⇨ RTS

The connectors at all PTS color systems are implemented in the same way.

3.2 Start-up

Attention! The correct operation of the PTS color systems are described in separate **operation manuals**.
Read the relevant chapters for these modules in the operation manual and read the safety precautions in chapter [1] carefully, before you power-on the Systems and before you steer the Systems by serial commands.

3.2.1 Control of PTS color system

Serial connector: RS232 interface

Parameters: 4800 ... 115200 baud, no parity, 1 stop bit, 8 data bit, no handshake

4. Command Syntax

For explanation of the command syntax, refer to the following table showing the meta characters and their meanings:

Syntax	Interpretation
<...>	The entry within the pointed brackets explains the value which is to be substituted in the destination INCLUDING the brackets. For example: <Current in A> would be, for a current of 28.5 Ampere replaced by 28.5
<CR>	Line closure sign CARRIAGE-RETURN with ASCII-Code 0D hex (13 decimal)
[...]	Square brackets contain optional information. This may be omitted if necessary

The additional indication ' - - -' is a repetition sign and shows that the part of the command which lay before in the square brackets may be repeated any desired number of times.

Control over the serial line interface is accomplished by use of command strings. These are defined as follows:

Command String = [<Prefix>.]<Command>[;<Command>[;….]]<CR>

As Perfix we must define r. or R. for commands for the reference standard and s. or S. for commands for the source.

Also we can define the standard Prefix with the command PCD (see capter 5.10).

The Maximum number of characters which may be used in a command string is 190, and this number must not be exceeded in any event!

The command will only start to be interpreted when a line termination character is received:

Termination character = <CR> or <;>

The Command is assembled as follows:

Command = <Command Code>[<1. Parameter>][,<further Parameter> - - -]

In the case of an error the commands generally will give back the complete command string with the extension '=E'.

e.g. U10,24.3<CR> → U10,24.3=E<CR> or SET<CR> →SET=E<CR>

If there is more than one command in a line, separated by semicolon (;), each command will give his own return line back.

e.g.: U1,230;U2,2300;U3,230<CR> → U1,230=O<CR>
U2,2300=E<CR>
U3,230=O<CR>

The output format of the return to a command depends on the actually set mode.

MODE=0 **Output format 0:** compatibility mode for older EMH equipment, the command is not repeated.

MODE=1 **Output format 1:** Extended mode, the command is repeated with the return.

For control of the output format see description to the MODE command in chapter 5.9.

Notation used in tables and command description

<CR> The carriage return <CR> at the end of each command and result string is not shown to simplify the notation.
Exception: <CR> at the beginning or within command strings are shown.

<O/E> Return to a command in output format 1 (MODE=1):
O: OK, command was accepted by the instrument
E: Error, command was not accepted

5. Command description for control of the internal reference standard

5.1 Readout of results

5.1.1 Results administration

The internal PTS software makes all measured results (instantaneous values, error values, etc.) available in a result list. For each result type an own place is reserved in the result list. The last measured value of this type is always placed at this position. If a new value is entered in the result list, then the old value is overwritten.

There are several methods to readout the results from the result list:

- Explicite readout of values of a result type
- Readout of the next not yet read value from the result list or readout of a result of high priority.
- Automatic readout of the next not yet read result or event = periodic simulation of the result request command (for test purposes with a terminal program).

The results can be enabled or disabled with the command **SP**.
With the commands **?<No>** single results: voltages, currents, power values etc. can be explicitly readout. All results are given back in character format with a fixed number of characters.
The command **SU1** starts an automatic result output from the instrument over the serial interface. All results that are enabled by the **SP** command are sent out over the serial interface as soon as new results are available in the result list.
The command **SU0** stops the automatic result output.
Additional to these commands, almost all commands that are used for changing settings and programming can also be used to read the actual settings, if they are sent without parameter.
A detailed description of the available results can be found in the following sections.

5.1.2 Types of results

There are two types of result classes to be distinguished:

- **Results within the result list:** The values can be readout explicitly with the command **?<No>**. The automatic check for new values in the result list can be done with the command **SP**. The individual results in the result list can be enabled or disabled for readout. This is done with the command **SP<No>**
- **Results with high priority:** These results cannot be readout explicitly. They are automatically generated by the system and they are treated with preference at the next readout of results.

Table 5-1 Result list

If not especially mentioned, the values in the result strings **<value>** have 8 characters filled up with spaces if necessary. If a value is below the starting value or out of range, 8 lines are displayed
'--------'. The result string starts always with an **E**. The second character represents the identification **<id>** of the result. All result strings are terminated with a carriage return **<CR>**.

If the result is not a new value, this is marked with the MSB set to 1 in the identification sign:

<id*> = <id> OR $80. By this operation <id*> shows normally a special sign, not the standard sign related to the corresponding result. In this way already read old values can be recognized by a program.

Result ?<No>	Flag SP<No>	Results	Result-String E<id>[,<value>[,<value>…]]
1	1	**Phase currents** I1, I2, I3 [A]	E@,<I1>,<I2>,<I3>
2	2	**Phase to neutral voltages** U1 U2, U3 [V]	EA,<U1>,<U2>,<U3>
3	3	**Active power per phase** P1, P2, P3 [W]	EB,<P1>,<P2>,<P3>
4	4	**Reactive power per phase** Q1, Q2, Q3 [var]	EC,<Q1>,<Q2>,<Q3>
5	5	**Apparent power per phase** S1, S2, S3 [VA]	ED,<S1>,<S2>,<S3>
6	6	**Active sum-power** ΣP [W]	EE, <ΣP>
7	7	**Reactive sum-power** ΣQ [var]	EF, <ΣQ>

Result ?<No>	Flag SP<No>	Results	Result-String E<id>[,<value>[,<value>…]]
8	8	**Apparent sum-power** ΣS [VA]	EG, <ΣS>
9	9	**Phase angle current to voltage** ϕui1, ϕui2, ϕui3 [°], ps ϕui1: phase angle I1 to U1 ps: phase sequence	EH,< ϕui1>,<ϕui2>,<ϕui3>,<ps> Phase sequence <ps>: <L>: Left <R>: Right <?>: Unknown
10	10	**Energy error** Error [%] of input 1 Error: Not enough impulses! (Message comes once, no result, error measurement is stopped) New number of impulses defined by the system (during mode error with time base TF)	EI,<E1> Eg EY,<imp> Formats of <E1>: < .->: Reset, initialisation < --.-->: Start-impulse <#####.##>: 2 digits after decimal point, positive error <-####.##>: 2 digits after decimal point, negative error <####.###>: 3 digits after decimal point, positive error <-###.###>: 3 digits after decimal point, negative error
12	12	**Phase to phase voltages** U12, U23, U31 [V]	EK,<U12>,<U23>,<U31>
13	13	**Signal frequency** frq [Hz], signal frequency	EL,<frq>
16	16	**Active, reactive and apparent sum power** <Psec>,<Qsec>,<Ssec>	EO, <Psec>,<Qsec>,<Ssec>
17	18	**U-samples during FFT-modes** FFT1, FFT4: U1 FFT2, FFT5: U2 FFT3, FFT6: U3 128 x 16bit samples of one period	EZ,<256 ASCII-HEX characters> 1 sample = 2 ASCII-HEX characters with 4 HEX signs for a 16bit-value: <16bit > = <low byte><high byte> <byte> = <high HEX><low HEX>
18	19	**I-samples during FFT-modes** FFT1, FFT4: I1 FFT2, FFT5: I2 FFT3, FFT6: I3 128 x 16bit samples of one period	E[,<256 ASCII-HEX characters> 1 sample = 2 ASCII-HEX characters with 4 HEX signs for a 16bit-value: <16bit > = <low byte><high byte> <byte> = <high HEX><low HEX>
19	20	**FFT-values (harmonics)** FFT1 … FFT3: U1 … U3 FFT4 … FFT6: I1 … I3 64 x 16bit FFT-values	E\,<128 ASCII-HEX characters> 1 FFT-value = 2 ASCII-HEX characters with 4 HEX signs: <16bit > = <low byte><high byte> <byte> = <high HEX><low HEX>
20	21	**Distortion during FFT modes** <distortion x> [%] FFT1 … FFT3: U1 … U3 FFT4 … FFT6: I1 … I3	E],<distortion x>
21	22	**Phase angles between voltages** ϕu1u2, ϕu2u3, ϕu3u1 [°] ϕu1u2: phase angle U1 to U2	E^,<ϕu1u2><ϕu2u3><ϕu3u1>
22	23	**Phase angles between currents** ϕi1i2, ϕi2i3, ϕi3i1 [°] ϕi1i2: phase angle I1 to I2	E_,<ϕi1i2><ϕi2i3><ϕi3i1>
23	24	**Energy and error of input 1** W1 [Wh/imp, varh/imp], W1: If the test period is n = 1 the measured constant of impulse input 1 is indicated. For n > 1 W1 shows the energy of one measurement period. In this case the constant of the impulse input can be calculated by dividing W1 with n. E1 [%]: Energy error measured with inpulse input 1	Ea,<W1>,<E1>
24	25	**Resting test duration during error measurements** The impulses in mode TU or the time in mode TF are counted down during the error measurement. restx Resting test duration of input x, x = 1,2,3 Test duration defined in impulses (TU): Resting impulses to new result are shown Test duration defined in seconds (TF): Resting time to new result is shown New values with interval = 0.2s, independent of the setting of T.	Eb,<rest1>,<rest2>,<rest3>

Result ?<No>	Flag SP<No>	Results	Result-String E<id>[,<value>[,<value>…]]
26	27	**Energy and error of input 2** W2 [Wh, varh], E2 [%] W2: Energy measured over one impulse of device under test at impulse input 2 E2: Energy error measured with impulse input 2	Ed,<W2>,<E2>
27	28	**Energy register values in [Ws] all seconds** WP+: active energy positive WP-: active energy negative WQ+: reactive energy positve WQ-: reactive energy negative New values with interval = 1s, independent of the setting of T.	Ee,<WP+>,<WP->,<WQ+>,<WQ-> values measured since last power up of the unit
28	29	**Energy register values in [kWh] all seconds** WP+: active energy positive WP-: active energy negative WQ+: reactive energy positve WQ-: reactive energy negative New values with interval = 1s, independent of the setting of T.	Ef,<WP+>,<WP->,<WQ+>,<WQ-> values measured since last power up of the unit
29	30	**Energy and error of input 3** W3 [Wh, varh], E3 [%] W3: Energy measured over one impulse of device under test at impulse input 3 E2: Energy error measured with impulse input 3	Eg,<W3>,<E3>
30	31	**Statistical data of input 1** S1: status of input 1, counting of the measured errors since last start of error measurement 0 = Waiting for start impulse 1 = First measurement running 2 .. x = Error value 1 .. x-1 N1: Number of repetitions regarded for calculation of mean value and standard deviation, counted up from 1 to N1, then remaining at N1 Em1: mean value of input 1, calculated over indicated repetitions N1 Ex1: standard deviation of input 1, calculated over indicated reptetitions N1 f: frequency of input 1	Eh,<S1>,<N1>,<Em1>,<Ex1>,<f1>
31	32	**Statistical data of input 2**, description see 30	Ei,<S2>,<N2>,<Em2>,<Ex2>,<f2>
32	33	**Statistical data of input 3**, description see 30	Ej,<S3>,<N3>,<Em3>,<Ex3>,<f3>
34	35	**Absolute phase angles** ϕu1r, ϕu2r, ϕu3r, ϕi1r, ϕi2r, ϕi3r [°] r: virtual reference All phase angles between U, I can be calculated based on these values (eg. ϕui1 = ϕu1r - ϕi1r)	El,<PhiAbsU1>,<PhiAbsU2>,<PhiAbsU3>,<PhiAbsI1>,<PhiAbsI2>,<PhiAbsI3>
36	37	**Actual output frequencies all seconds** OUT1, OUT2, OUT3 [Hz] New values with interval = 1s, independent of the setting of T.	En,<fout1>,<fout2>,<fout3>

Table 5-2 Results with high priority

The results cannot be locked. There are no flags to disable high priority results. The SP command has no influence. Results of this kind are indicating termination of a function or a warning, if the status of the unit has changed.

High priority result	Result-String E<id>[,<value>[,<value>…]]
Range overflow Generated in 1s interval as long as there is an overflow.	EQ
End of time base measurement Immediately followed by new values	ER
New ranges switched on	ES

5.1.3 Readout of results (SP,?,SU,RDR,RDRBU,RDRBI,RDRCC,RDRCT,RDRF,RDRFT, RDRM,RDRTI,RDRTU,RSI,RSU,RWI,RWU)

Cmd	Input format	Output format 1	Description
SP	**SP<No>**	**SP<No>=<flag>**	**Read <flag> status of result <No>**
	SP<No>,<flag>	**SP<No>,<flag>=<O/E>**	**Set <flag> of result <No>**

<No> Flag number of result regarding result list, see table 5-1.
Note: The flag <No> for results 17 … 36 is different than the result number: <No> = result number + 1!
0: All results (set command only)

<flag> 0: Disable result in result list
1: Enable result in result list
2: Enable result exclusively in result list (all other results are disabled, selected result is enabled)

SP0,0	SP0,0=O	Disable all results in the result list
SP0,1	SP0,1=O	Enable all results in the result list
SP10,2	SP10,2=O	Enable exclusively error E1 in result list

SU	**SU<0/1>**	**SU<0/1>=<O/E>**	**Start / Stop automatic sending of results**

<0/1> 1: Start internal check for new results with simulation of **?** command (10 ms interval). New results, which are enabled in the result list, are send out as soon as they are available.
0: Stop automatic sending of results.

Note: This command is forseen for test purposes with an ASCII terminal program (e.g. windows hyper terminal). It is not recommended, to use this command for controlled readout of a results in a program, because the allocation between a request and a received result is not clear.

SP6,2	SP6,2=O	Enable exclusively active power ΣP in result list.
T1	T1=O	Set time base 1s.
SU1	SU1=O	Start automatic sending of results. The result list is checked for new results in 10ms intervals.
	ER<CR> EE,2988.605	Time base interval finished, new results available New result No. 6 available: ΣP = 2988.605 W
	ER<CR> EE,2988.763	Time base interval finished, new results available New result No. 6 available: ΣP = 2988.763 W
SU0	SU0=O	Stop automatic sending of results.

?	**?<No>**	**E<id>[,<value>[,<value>…]]**	**Readout of result <No>**

<No> Result number regarding result list, see table 5-1

<id> Identification sign for new value, see Tab 5-1, if the result is not a new value, this is marked with the MSB set to 1 in the identification sign: <id*> = <id> OR $80. For description of result strings see table 5-1.

RDR	**RDR**	**RDR= E<id>[,<value>[,<value>…]]**	**All momentary values and events read out at the same time**

The readings of the RDR command are simmular than the results of the ? commands. (see chapter 5.1.2)

RDRBI	**RDRBI**	**RDRBI= <value>[,<value>…]**	**Results of current burden measurements**

Result Nr.	Typ	Data	Description
1	UBYTE	Phase	Phase 1…3 → Block is invalid
2	FLOAT	In	Nominal current (A)
3	FLOAT	SN	Nominal Burden in (VA)
4	FLOAT	l	Length of the cables (m)
5	FLOAT	A	Cross section of the cables (mm2)
6	FLOAT	U	Voltage (V)
7	FLOAT	I	Current (A)
8	FLOAT	R	Active recistance (Ω)
9	FLOAT	jX	Reactive recistance (Ω)
10	FLOAT	Z	Apprent recistance (Ω)
11	FLOAT	Sb	Actual burden (%)
12	FLOAT	Sn	Apprent power at nominal conditions
13	FLOAT	Sns	Apprent power incl. conduction losses at nominal conditions
14	FLOAT	cosβ	Power factor cos Beta

Cmd	Input format	Output format 1	Description
RDRBU	**RDRBU**	**RDRBI= <value>[,<value>…]**	**Results of voltage burden measurements**

Result Nr.	Typ	Data	Description
1	UBYTE	Phase	Phase 1…3 → Block is invalid
2	FLOAT	Un	Nominal voltage (V)
3	FLOAT	SN	Nominal Burden in (VA)
4	FLOAT	RF	Fuse resistance (Ω)
5	FLOAT	L	Length of the cables (m)
6	FLOAT	A	Cross section of the cables (mm2)
7	FLOAT	U	Voltage (V)
8	FLOAT	I	Current (A)
9	FLOAT	G	Conductance (S)
10	FLOAT	jB	Susceptance (S)
11	FLOAT	Y	Admittance (S)
12	FLOAT	Sb	Actual burden (%)
13	FLOAT	Sn	Apprent power at nominal conditions
14	FLOAT	Sns	Apprent power incl. conduction losses at nominal conditions
15	FLOAT	cosβ	Power factor cos Beta

Cmd	Input format	Output format 1	Description
RDRCC	**RDRCC**	**RDRCC= <value>[,<value>…]**	**Results of last meter constant evaluation**

Result Nr.	Typ	Data	Description
1	BYTE	Reference [3]	That's the power mode which was reference for the calculation of the meter constant. This value is identical with the referencevalue of the command DFO. Exeption: If the Input is inactive the response is -1 Example: -1 = off 10 = P1 19 = PSum 20 = QSum 21 = SSum
2	UBYTE	Mode[3]	Measuring Mode Mode = 0: interval in Impulses Mode = 1: Interval in ms
3	ULONG	Interval[3]	Measuringinterval in Impulses or ms.
4	UBYTE	CPUnit[3]	The value of the meterconstant CPUnit = 0: Value = i/k?h, CPUnit = 1: Value = i/?h, CPUnit = 2: Value = i/?s, CPUnit = 3: Value = k?h/i CPUnit = 4: Value = ?h/i CPUnit = 5: Value = ?s/i
5	FLOAT	CP[3]	Calculated Meterconstant in i/?s

Cmd	Input format	Output format 1	Description
RDRCT	**RDRCT**	**RDRCT= <value>[,<value>…]**	**Results of last register test**

Result Nr.	**Typ**	**Data**	**Description**
1	BYTE	Reference [3]	That's the power mode which was reference for the register test. This value is identical with the referencevalue of the command DFO Exeption: If the Input is inactive the response is -1 Example: -1 = off 10 = P1 19 = PSum 20 = QSum 21 = SSum
2	FLOAT	Interval [3]	Measuringinterval in minutes
3	FLOAT	EMin [3]	Lower error limit
4	FLOAT	EMax [3]	Upper error limit
5	FLOAT	Energy [3]	Dosage energy in k?h
6	UBYTE	EnUnit [3]	The value of the input for the dosage energy EnUnit = 0: Einheit = ?s EnUnit = 1: Einheit = ?h EnUnit = 2: Einheit = k?h EnUnit = 3: Einheit = M?h EnUnit = 4: Einheit = G?h
7	UBYTE	nVKDig [3]	Number of digits before the comma
8	UBYTE	nNKDig [3]	Number of digits behind the comma
9	FLOAT	W [3]	Measured energy since the start of the measurement
10	FLOAT	WMax [3]	Measured maximum power since the start of the measurement
11	FLOAT	Cntr1 [3]	Counter value before the register test
12	FLOAT	Cntr2 [3]	Counter value after the register test
13	FLOAT	Error [3]	Error in %
14	FLOAT	tm [3]	Time of the signal periode in seconds (s)
15	FLOAT	te [3]	Disconnecting time in seconds (s)

Cmd	Input format	Output format 1	Description
RDRF	**RDRF**	**RDRF= <value>[,<value>…]**	**Results of last error measurement**

Result Nr.	Typ	Data	Description
1	BYTE	Reference [3]	That's the power mode which was reference for the error measurement. This value is identical with the referencevalue of the command DFO Exeption: If the Input is inactive the response is -1 Example: -1 = off 10 = P1 19 = PSum 20 = QSum 21 = SSum
2	UBYTE	CPUnit[3]	The value of the meterconstant CPUnit = 0: Value = i/k?h, CPUnit = 1: Value = i/?h, CPUnit = 2: Value = i/?s, CPUnit = 3: Value = k?h/i CPUnit = 4: Value = ?h/i CPUnit = 5: Value = ?s/i
3	UBYTE	Mode[3]	Measuring Mode Mode = 0: interval in Impulses Mode = 1: Interval in ms
4	ULONG	Interval[3]	Measuringinterval in Impulses or ms.
5	FLOAT	CP [3]	Constant of the meter under test in i/?s
6	FLOAT	EMin [3]	Lower error limit
7	FLOAT	EMax [3]	Upper error limit
8	ULONG	N [3]	Parameterize numbers of values in ErrorEs and ErrorEs
9	FLOAT	Error [3]	The last measured error
10	ULONG	ErrN [3]	Numbers of values in ErrorEs and ErrorEs
11	FLOAT	ErrorEm	Mean value calculated by paramterize numbers (N) of error measurements
12	FLOAT	ErrorEs	Standart deviation calculated by paramterize numbers (N) of error measurements

Cmd	Input format	Output format 1	Description
RDRM	**RDRM**	**RDRM= <value>[,<value>…]**	**Results of last energy and maximum register test**

Result Nr.	Typ	Data	Description
1	BYTE	Reference [3]	That's the power mode which was reference for the last energy measurement. This value is identical with the referencevalue of the command DFO Exeption: If the Input is inactive the response is -1 Example: -1 = off 10 = P1 19 = PSum 20 = QSum 21 = SSum
2	FLOAT	Interval[3]	Measuringinterval in Minutes
3	FLOAT	W [3]	Measured energy since the start of the measurement
4	FLOAT	WMax [3]	Measured maximum power since the start of the measurement

Cmd	Input format	Output format 1	Description
RDRTI	**RDRTI**	**RDRTI= <value>[,<value>…]**	**Results of last current transformer ratio measurement**

Result Nr.	Typ	Data	Description
1	UBYTE	PrimInput	Primary measuring input 0 = high current outputs (up to 120A) 1 = low current outputs (up to 12A) 2 = 100A current clamps 3 = 1000A current clamps 4 = 30A Flex 5 = 300A Flex 6 = 3000A Flex 7 = Current Hotsticks
2	UBYTE	SekInput	Secondary measuring input 0 = high current outputs (up to 120A) 1 = low current outputs (up to 12A) 2 = 100A current clamps 3 = 1000A current clamps 4 = 30A Flex 5 = 300A Flex 6 = 3000A Flex 7 = Current Hotsticks
3	UBYTE	RefFlag	This result select if the primary or the secondary value is the base to calculate the nominal values (NP/NS). 0 = Calc NS, Ref NP 1 = Calc NP, Ref NS
4	FLOAT	NPNom [3]	Primary nominal transformer values for phase 1 to 3
5	FLOAT	NSNom [3]	Secondary nominal transformer value for phase 1 to 3
6	FLOAT	Ip [3]	Measured primary current in A
7	FLOAT	Is [3]	Measured secondary current in A
8	FLOAT	Phi [3]	Phaseangle between primary and secondary winding in degrees for phase 1 to 3
9	FLOAT	NPcalc	Calculated primary values for phase 1 to 3
10	FLOAT	NScalc	Calculated secondary values for phase 1 to 3
11	FLOAT	Error [3]	Transformer error in % for phase 1 to 3

Cmd	Input format	Output format 1	Description
RDRTU	**RDRTU**	**RDRTU= <value>[,<value>…]**	**Results of last voltage transformer ratio measurement**

Result Nr.	**Typ**	**Data**	**Description**
1	UBYTE	PrimInput	Primary measuring input 0 = direct voltage inputs 8 = Voltage Hotsticks
2	UBYTE	SekInput	Secondary measuring input 0 = direct voltage inputs 8 = Voltage Hotsticks
3	UBYTE	RefFlag	This result select if the primary or the secondary value is the base to calculate the nominal values (NP/NS). 0 = Calc NS, Ref NP 1 = Calc NP, Ref NS
4	FLOAT	NPNom [3]	Primary nominal transformer values for phase 1 to 3
5	FLOAT	NSNom [3]	Secondary nominal transformer value for phase 1 to 3
6	FLOAT	Ip [3]	Measured primary voltage in V
7	FLOAT	Is [3]	Measured secondary voltage in V
8	FLOAT	Phi [3]	Phaseangle between primary and secondary winding in degrees for phase 1 to 3
9	FLOAT	NPcalc	Calculated primary values for phase 1 to 3
10	FLOAT	NScalc	Calculated secondary values for phase 1 to 3
11	FLOAT	Error [3]	Transformer error in % for phase 1 to 3

Cmd	Input format	Output format 1	Description
RSI	**RSI<ph>**	**RSI<ph>=<nf>,<re0><im0> … <re31><im31>**	**Read current harmonics of one phase**

RSU	**RSU<ph>**	**RSU<ph>=<nf>,<re0><im0> … <re31><im31>**	**Read voltage harmonics of one phase**

<ph> Phase L1, L2, L3: ph = 1, 2, 3

<nf> New Flag: 0: No new values since last readout, 1: New values,
The values in the PTS color system are periodically updated in the interval of the time base T.

<rex>,<imx> Real part (re), imaginary part (im) of harmonic x, x = 0 … 31,
0: DC component, 1: Fundamental, 2 .. 31: 2nd to 31st harmonic
Format: String = (+/-)Z.zzzzE(+/-)Z (10 characters)

VALUE [V,A] = (<value> / 32767) * FS [V,A]

with <value> = <rex> or <imx>, x = 0 … 31, of U,I and FS = Full Scale value of actual selected range of U or I.
Use commands I, DB and table 5-3 or U, DBU and table 5-4 to find actual FS value of I or U.

RSI2	RSI2=1,+1.2997E+1+0.0000E+0 +4.6059E+3+1.2853E+4 …	Readout harmonics of current I2, new values available (only re, im of harmonics 0 and 1 shown)
RSI2	RSI2=0,+1.2997E+1+0.0000E+0 +4.6059E+3+1.2853E+4 …	Readout harmonics of current I2, <nf> = 0, old values (only re, im of harmonics 0 and 1 shown)

RWI	**RWI<ph>**	**RWI<ph>=<nf>,<s0><s1> … <s127>**	**Read current wave form of one phase**

RWU	**RWU<ph>**	**RWU<ph>=<nf>,<s0><s1> … <s127>**	**Read voltage wave form of one phase**

<ph> Phase L1, L2, L3: ph = 1, 2, 3

<nf> New Flag: 0: No new values since last readout, 1: New values
The values in the PRS 300.3 module are periodically updated in the interval of the time base T.

<s0> … <s127> Values of 128 samples of one signal period, Format: String = (+/-)Z.zzzzE(+/-)Z (10 characters)

VALUE [V,A] = (<sx> / 32767) * FS [V,A] * √2

with <sx>, x = 0 … 127, of U,I and FS = Full Scale value of actual selected range of U or I.
Use commands I, DB and table 5-3 or U, DBU and table 5-4 to find actual FS value of I or U.

RWU1	RWU1=1,-1.9357E+4 -1.8726E+4 …	Readout wave form of voltage U1, new values available (only two samples shown)
RWU1	RWU1=0,-1.9357E+4 -1.8726E+4 …	Readout wave form of voltage U1, <nf> = 0, old values (only two samples shown)

5.2 General settings (R,T,I,U,DB,DBU,F,DFO,DFM,AUT,MAN,YI,YU,IAL,UAL)

Cmd	Input format	Output format 1	Description
R	**R**	**<CR>PTSx.3C**	**Reset**

Reset of the system. The default settings are activated: P4, DB0, SP0, SU0, AUT, MODE0. The system is in the same status as after power-on. After reset the name of the system is shown

R	R=O<CR><CR>PTSx.3C	Reset at MODE=1, after reset MODE=0 (output format 0) is activated

T	**T**	**T=<time>**	**Read time base**
	T<time>	**T<time>=<O/E>**	**Set time base**

<time> Time base in [s], interval for new results, range: 0.1 … 999.9 s or
0: external synchronization by impulse input 1, string = 5 chars right aligned with leading blanks

T	T= 1	Read actual time base, T = 1s (string = 4 x space + '1')
T2	T2=O	Set time base 2s, O: OK, value is accepted.
T10000	T10000=E	Set time base 10000s, E: error, value out of range, value is not changed.
T0	T0=O	Set 0s, ext. synch. of T by impulses at impulse input 1. (Error Measurement on Input 1 isn't possible)

I	**I**	**I=<ri1><ri2><ri3>**	**Read current ranges**
	I<ri1>,<ri2>,<ri3>	**I<ri1>,<ri2>,<ri3>=<O/E>**	**Set current ranges**

<rix> Current range number of phase x, x = 1,2,3
1: Lowest range, the numbering of the ranges depends on the actually used input <ini>, details regarding range numbering see table 5-3.

Note: New settings of ranges become only active at MAN range selection. During AUT range selection they are ignored.

DB	DB=1	12A input active
I	I=221	I1, I2 are measured in current range 2, I3 is measured in current range 1
I3,3,3	I3,3,3=O	Set current range 3 for all phases, O: OK, new range settings are accepted.

U	**U**	**U=<ru1><ru2><ru3>**	**Read voltage ranges**
	U<ru1>,<ru2>,<ru3>	**U<ru1>,<ru2>,<ru3>=<O/E>**	**Set voltage ranges**

<rux> Voltage range number of phase x, x = 1,2,3
1: Lowest range, the numbering of the ranges depends on the actually used input <inu>, details regarding range numbering see table 5-4.

Note: New settings of ranges become only active at MAN range selection. During AUT range selection they are ignored.

DBU	DBU=0	Standard voltage input active
U	I=221	U1, U2 are measured in voltage range 2, U3 is measured in voltage range 1
U3,3,3	U3,3,3=O	Set voltage range 3 for all phases, O: OK, new range settings are accepted.

DB	**DB**	**DB=<ini>**	**Read current input selection**
	DB<ini>	**DB<ini>=<O/E>**	**Select current input <ini>**

<ini> Current input number

<ini>	**Input**
0	Standard current input (120A) (default setting)
1	Alternativ current input (12A)
2	Clamp-on CT100A
3	Clamp-on CT1000A
4	LEMflex 30A
5	LEMflex 300A
6	LEMflex 3000A
7	Hotstick U connected to current path
8	Hotstick I

<dbi>=7 is used to redirect the voltage measurement to the current path. This is used during ratio test of voltage transformers to measure primary and secondary voltage at the same time.

DB	DB=0	Read status, standard current input selected
DB2	DB2=O	Select current input 2, clamp-on CT100A

Cmd	Input format	Output format 1	Description
DBU	**DBU**	**DBU=<inu>**	**Read voltage input selection**
	DBU<inu>	**DBU<inu>=<O/E>**	**Select voltage input <inu>**

<inu> Voltage input number

<inu>	**Input**
0	Standard voltage input (default setting)
1	Alternative voltgage input
2	Clamp-on CT100A connected to voltage path
3	Clamp-on CT1000A connected to voltage path
4	LEMflex 30A connected to voltage path
5	LEMflex 300A connected to voltage path
6	LEMflex 3000A connected to voltage path
7	Hotstick U
8	Hotstick I connected to voltage path

<inu> = 1 is not existing at PRS 400.3
<inu> = 2 … 6 and 8 are used to redirect the current measurement to the voltage path. This is used during ratio test of current transformers to measure primary and secondary current at the same time.

DBU	DBU=0	Read status, standard voltage inputs selected
DBU7	DBU7=O	Select voltage input Hotstick U

F	**F**	**F=<ref>**	**Read reference of impulse output 1**
	F<ref>	**F<ref>=<O/E>**	**Set reference of impulse output 1**

<ref> Reference number for impulse output 1

<ref>	**Reference**
1	P
2	Q
3	S
4	1000 imp. / revolution (not supported by PRS 400.3)
5	I^2*R I^2*R = ($I1^2$ + $I2^2$ + $I3^2$) * R, with R = 1Ω
6	U^2/R U^2/R = ($U1^2$ + $U2^2$ + $U3^2$) / R, with R = 1MΩ

P4	P4=O	Set active 4 wire mode
F1	F1=O	Set fout1 proportional to P
F	F=1	Read reference of fout1, reference is P
K4	K4=O	Set artificial reactive 4 wire mode
F3	F3=O	Set fout1 proportional to S

DFO	**DFO<out>**	**DFO<out>=<x>,<c>**	**Read reference, constant of impulse output**
	DFO<out>,<x>[,<c>]	**DFO<out>,<x>[,<c>]=O**	**Set reference, constant of impulse output**

<out> Impulse output number

<out>	**Impulse generator**
empty	Status of impulse output is returned
1	Impulse output 1

<x> Reference value for impulse output, see table 5-5

<c> Optional, definition of fix meter constant in [i/Ws] for impulse output, independent of internal ranges, 0: standard, range dependent constant regarding technical data (default)

DFO1	DFO1=34,0	Read status of impulse output 1: PΣ4, 0:default constant, range dependent
DFO	DFO1=34,0 DFO2=38,100 DFO3=32,0	Read status of all impulse outputs: Out1: PΣ4, default constant Out2: QNΣ4, constant = 100 i/vars Out3: SΣ4, default constant
DFO1,34,0.00277778	DFO1,34,0.00277778=O	Set Out1: PΣ4, 10’000 imp/kWh=0.00277778 i/Ws

DFM	**DFM<ImpGen>**	**DFM<ImpGen>,<mode>**	**Read energy direction of the output frequency**
	DFM<ImpGen>,<mode>	**DFM<ImpGen>,<mode>=<O/E>**	**Set energy direction of the output frequency**

<Mode> Energy direction (+/-)
1 = The impulse output is independent of the energy direction
2 = The impulse output is only active when the reference values are positive (+)
3 = The impulse output is only active when the reference values are negative (-)

<ImpGen> Impulse output number

<ImpGen>	**Impulse generator**
1	Impulse output 1

Cmd	Input format	Output format 1	Description
AUT	**AUT**	**AUT=<O/E>**	**Set automatic range selection**

The automatic range selection for internal current ranges is activated. The actual settings for range synchronization (YI, YU) and lowest range limits (IAL, UAL) are regarded.
The actual selected ranges can be readout with the commands I, U, SG.

MAN	**MAN**	**MAN=<O/E>**	**Set manual range selection**

The manual range selection for internal current ranges is activated. The actual settings for range synchronization (YI, YU) are regarded. Range settings with commands I, U, defined during AUT mode, are now activated.

MAN	MAN=O	Activate manual range selection.
YI0	YI0=O	Disable current range synchronization
I2,3,4	I2,3,4=O	Set ranges 2, 3, 4 for phases 1, 2, 3
I	I=234	Actual ranges of phases 1, 2, 3 are 2, 3, 4
YI1	YI1=O	Enable current range synchronization
I	I=444	Highest range 4 is set for all phases

YI	**YI**	**YI=<0/1>**	**Read status of current range synchronisation**
	YI<0/1>	**YI<0/1>=<O/E>**	**Set current range synchronisation**

<0/1> 0: Current range synchronization disabled. At automatic (AUT) range selction, the ranges are selected individual per phase
1: Current range synchronization enabled. At automatic (AUT) and manual (MAN) range selection the highest range is used for all phases.

YI	YI=0	Status, current range synchronization is disabled
AUT	AUT=O	Activate automatic range selection
I	I=3,1,1	Read current ranges, only phase 1 measures in range 3, phases 2, 3 are measuring in range 1.
YI1	YI1=O	Activate current range synchronization
I	I=3,3,3	All phases are measuring in the same range 3.

YU	**YU**	**YU=<0/1>**	**Read status of voltage range synchronisation**
	YU<0/1>	**YU<0/1>=<O/E>**	**Set voltage range synchronisation**

<0/1> 0: Voltage range synchronization disabled. At automatic (AUT) range selction, the ranges are selected individual per phase
1: Voltage range synchronization enabled. At automatic (AUT) and manual (MAN) range selection the highest range is used for all phases.

YU	YU=0	Status, voltage range synchronization is disabled
AUT	AUT=O	Activate automatic range selection
U	u=3,1,1	Read voltage ranges, only phase 1 measures in range 3, phases 2, 3 are measuring in range 1.
YU1	YU1=O	Activate current range synchronization
U	U=3,3,3	All phases are measuring in the same range 3.

IAL	**IAL**	**IAL=<0/1>**	**Read status of current range limit for AUT**
	IAL<0/1>	**IAL<0/1>=<O/E>**	**Set current range limit for AUT**

<0/1> 0: Lower current range limit for automatic range selection (AUT) is disabled. All current ranges can be selected.
1: Lower current range limit is enabled. AUT does not select current ranges

IAL	IAL=0	Status, current range limit for AUT disabled
IAL1	IAL1=O	Enable current range limit for AUT

UAL	**UAL**	**UAL=<0/1>**	**Read status of voltage range limit for AUT**
	UAL<0/1>	**UAL<0/1>=<O/E>**	**Set voltage range limit for AUT**

<0/1> 0: Lower voltage range limit for automatic range selection (AUT) is disabled. All current ranges can be selected.
1: Lower voltage range limit is enabled. AUT does not select voltages ranges
E.g. limit is disabled during burden measurement of current transformers, where the lowest voltage ranges (1 = 0.4V, 1 = 5V) are needed.

AUT	AUT=O	Activate automatic range selection
UAL1	UAL1=O	Enable voltage range limit for AUT
U	U=333	Read voltage ranges, all phases in range 3
UAL0	UAL0=O	Disable voltage range limit for AUT
U	U=111	Read voltage ranges, all phases in range 1

Cmd	Input format	Output format 1	Description
TFI	**TFI**	**TFI=<TransfactorI1>, …..**	**Read out the actuell used current transformer factors**
	TFI=<TransfactorI1>, ….	**TFI=<TransfactorI1>,... = O**	**Set the current transformer factors**
	TFI0	**TFI=0**	**Reset the current transformer factors**

Example:

?1<CR> ; Read out the acual current values
E@, 1.000012, 2.000032, 3.000022
TFI<CR> ; Read out the actuell current transformer factor settings
TFI=0<CR>
TFI10,0.1,1<CR> ; Set the current transformer factors I1=10, I2=0.1, I3=1
TFI<CR> ; Read out the actuell current transformer factor settings
<whait one measuring periode>
?1<CR> ; Read out the acual current values
E@, 10.000012, 0.200003, 3.000022
TFI0<CR> ; Reset the current transformer factor settings
<whait one measuring periode>
?1<CR> ; Read out the acual current values
E@, 1.000012, 2.000032, 3.000022

TFU	**TFU**	**TFU=<TransfactorU1>, …..**	**Read out the actuell used voltage transformer factors**
	TFU=<TransfactorI1>, ….	**TFU=<TransfactorU1>,... = O**	**Set the voltage transformer factors**
	TFU0	**TFU=0**	**Reset the voltage transformer factors**

Example:

?2<CR> ; Read out the acual voltage values
E@, 230.0023, 230.0032, 230.0011
TFU<CR> ; Read out the actuell voltage transformer factor settings
TFU=0<CR>
TFU20,0.5,1<CR> ; Set the voltage transformer factors U1=20, U2=0.5, U3=1
TFU<CR> ; Read out the actuell voltage transformer factor settings
<whait one measuring periode>
?2<CR> ; Read out the acual voltage values
E@, 4600.002, 115.0032, 230.0011
TFU0<CR> ; Reset the voltage transformer factor settings
<whait one measuring periode>
?2<CR> ; Read out the acual voltage values
E@, 230.0023, 230.0032, 230.0011

Cmd	Input format	Output format 1	Description
TRI	**TRI**	**TRI1=<PrimCur1>,<SecCur1>** **TRI2=<PrimCur2>,<SecCur2>** **TRI3=<PrimCur3>,<SecCur3>**	**Read out the actuell used current ratios**
	TRI=<Phase>,<PirmCur>,<SecCur>	**TRI=<Phase>,<PirmCur>,<SecCur>=O**	**Set the current transformer ratios**
	TRI0	**TRI=0**	**Reset the current transformer ratios**

<PrimCur1> Primary transformer current phase 1
<SecCur1> Secondary transformer current phase 1

Example:

?1<CR> ; Read out the acual current values
E@, 1.000012, 2.000032, 3.000022
TRI<CR> ; Read out the actuell current transformer ratio settings
TRI=0<CR>
TRI1,10,1<CR> ; Set the current transformer ratio I1=10:1
TRI2,1,1<CR> ; Set the current transformer ratio I2=1:1
TRI3,100,5<CR> ; Set the current transformer ratio I3=100:5
TRI<CR> ; Read out the actuell current transformer factor settings
TRI1,10,1<CR>
TRI2,1,1<CR>
TRI3,100,5<CR>
<whait one measuring periode>
?1<CR> ; Read out the acual current values
E@, 10.000012, 2.000032, 60.00044
TRI0<CR> ; Reset the current transformer ratio settings
<whait one measuring periode>
?1<CR> ; Read out the acual current values
E@, 1.000012, 2.000032, 3.000022

TRU	**TRU**	**TRI1=<PrimVol1>,<SecVol1>** **TRI2=<PrimVol2>,<SecVol2>** **TRI3=<PrimVol3>,<SecVol3>**	**Read out the actuell used voltage transformer ratios**
	TRU=<Phase>,<PirmVol>,<SecVol>	**TRU=<Phase>,<PirmVol>,<SecVol>=O**	**Set the voltage transformer ratio**
	TRU0	**TRU=0**	**Reset the voltage transformer ratio settings**

<PrimVol1> Primary transformer voltage phase 1
<SecVol1> Secondary transformer voltage phase 1

Example:

?2<CR> ; Read out the acual voltage values
E@, 230.0023, 230.0032, 230.0011
TRU<CR> ; Read out the actuell voltage transformer ratio settings
TRU=0<CR>
TRU1,10,1<CR> ; Set the voltage transformer ratios U1=10:1
TRU2,1,1<CR> ; Set the voltage transformer ratios U2=1:1
TRU3,100,5<CR> ; Set the voltage transformer ratios U3=100:5
TRU<CR> ; Read out the actuell voltage transformer ratio settings
TRIU,10,1<CR>
TRU2,1,1<CR>
TRU3,100,5<CR>
<whait one measuring periode>
?2<CR> ; Read out the acual voltage values
E@, 2300.002, 230.0032, 4600.022
TRU0<CR> ; Reset the voltage transformer settings
<whait one measuring periode>
?2<CR> ; Read out the acual voltage values
E@, 230.0023, 230.0032, 230.0011

Table 5-3 Current range numbering for different inputs

Definition of current range numbering **<rix>** for phase **x = 1, 2, 3**, used with command **I**, for different inputs **<ini>**, used with command **DB**. To read out the parameters, work with the command **TPM** (chapter 5.8).

The table shows the end of range values in [A] for current inputs and in [V] for voltage inputs routed to the current path.

PTS2.3C / PTS3.1C (Internal reference meter SWS2.3)

		I range number <rix> for phase x = 1, 2, 3								
DB input	**<ini>**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**Parameters**
120A (default)	**0**	0.12	1.2	12	120					PAR1 ... PAR8
12A	**1**	0.012	0.12	1.2	12					(PAR1 ... PAR 8) / PAR102 [2]
CT100A	**2**	0.1	1	10	100					PAR301 ... PAR304
CT1000A	**3**	1	10	100	1000					PAR401 ... PAR404
LEM30A	**4**	30								PAR501
LEM300A	**5**	300								PAR502
LEM3000A	**6**	3000								PAR503

PTS3.3C (Internal reference meter PRS1.3)

		I range number <rix> for phase x = 1, 2, 3								
DB input	**<ini>**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**Parameters**
120A (default)	**0**	0.04	0.12	0.4	1.2	4	12	40	120	PAR1 ... PAR8
12A	**1**	0.004	0.012	0.04	0.12	0.4	1.2	4	12	(PAR1 ... PAR 8) / PAR102 [2]
CT100A	**2**	0.8	4	20	100					PAR301 ... PAR304

[2] The end of range values for the 120A input, defined in PAR1 ... PAR8, must be divided by the weighting factor PAR102 = 10 to get the end of range values for the 12A input.

Table 5-4 Voltage range numbering for different inputs

Definition of voltage range numbering **<rux>** for phase **x = 1, 2, 3**, used with command **U**, for different inputs **<inu>**, used with command **DBU**. To read out the parameters, work with the command **TPM** (chapter 5.8).

The table shows the end of range values in [V] for voltage inputs and in [A] for current inputs routed to the voltage path.

PTS2.3C / PTS3.1C (Internal reference meter SWS2.3)

		U range number <rux> for phase x = 1, 2, 3								
DBU input	**<inu>**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**Parameters**
U direct (def.)	**0**	0.5	5	250	500					PAR9 ... PAR16
U alternative [2]	**1**									
CT100A [1]	**2**	0.1	1	10	100					PAR301 ... PAR304
CT1000A [1]	**3**	1	10	100	1000					PAR401 ... PAR404
LEM30A [1]	**4**	30								PAR501
LEM300A [1]	**5**	300								PAR502
LEM3000A [1]	**6**	3000								PAR503

PTS3.3C (Internal reference meter PRS1.3)

		U range number <rux> for phase x = 1, 2, 3								
DBU input	**<inu>**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**Parameters**
U direct (def.)	**0**	0.4	5	60	120	240	480			PAR9 ... PAR16
U alternative [2]	**1**									
CT100A [1]	**2**	0.8	4	20	100					PAR301 ... PAR304

[1] Current path internally routed to voltage path, used for ratio test of current transformers, where primary and secondary current must be measured at the same time.
[2] Not existing at PTS color

Table 5-5 Reference values for impulse outputs and internal impulse generators

The **reference values** with the number **<x>** are used with the commands FOUT, DFO, FFSX, WRS.
<x> is used to define the reference values of the 6 available impulse generators:

- Impulse generators 1 to 3 = impulse outputs 1 to 3
- Three internal generators for error measurement (4), dosage (5) and energy measurement (6).
  Note: The internal generators 4, 5, 6 are not supported by the PTSx.xC. At other instruments the internal generators are not really generating impulses. They are reference for the internal registers used for the calculations of error, energy and dosage.

The table shows also the default setting for <f>, if this parameter is omitted in the FOUT command, and the impulse output constant CPZ.

<X>	Reference Value	Description	<f> (default)	CPZ [i/…s]
0	**Time**	Output of fix frequency: f = fmax * <f>, fmax = 58'500 Hz (PAR103)	1	fmax*f
1	**U1**	Phase to neutral voltage of phase 1	1	fmax / Un * f
2	**U2**	Phase to neutral voltage of phase 2	1	fmax / Un * f
3	**U3**	Phase to neutral voltage of phase 3	1	fmax / Un * f
4	**I1**	Current of phase 1	1	fmax / In * f
5	**I2**	Current of phase 2	1	fmax / In * f
6	**I3**	Current of phase 3	1	fmax / In * f
7	**U12**	Phase to phase voltage U12 = U1 - U2	0.5	fmax / Un * f
8	**U23**	Phase to phase voltage U23 = U2 - U3	0.5	fmax / Un * f
9	**U31**	Phase to phase voltage U31 = U3 - U1	0.5	fmax / Un * f
10	**P1**	Active power phase 1*	1	fmax / (Un * In) * f
11	**P2**	Active power phase 2*	1	fmax / (Un * In) * f
12	**P3**	Active power phase 3*	1	fmax / (Un * In) * f
13	**Q1**	Reactive power phase 1*	0.333333	fmax / (Un * In) * f
14	**Q2**	Reactive power phase 2*	0.333333	fmax / (Un * In) * f
15	**Q3**	Reactive power phase 3*	0.333333	fmax / (Un * In) * f
16	**S1**	Apparent power phase 1	1	fmax / (Un * In) * f
17	**S2**	Apparent power phase 2	1	fmax / (Un * In) * f
18	**S3**	Apparent power phase 3	1	fmax / (Un * In) * f
19	**PΣ**	Total active power*	0.333333	fmax / (Un * In) * f
20	**QΣ**	Total reactive power*	0.333333	fmax / (Un * In) * f
21	**SΣ**	Total apparent power*	0.333333	fmax / (Un * In) * f
22	**I^2*R**	Power of currents I at R=1Ω, with $I^2 = I1^2 + I2^2 + I3^2$	0.333333	fmax / In^2 * f
23	**U^2/R**	Power of voltages U(L-N) at R=1MΩ, with $U^2 = U1^2 + U2^2 + U3^{2*}$	333333	fmax / Un^2 * f
*		Depend on the actual settings of the measuring mode (3wire or 4wire) respectively of the reactive power mode (Qn or Qx)		

Note: Un, In are the Full Scale (FS) values of the actual selected highest voltage and current range.
Use the commands I, DB and table 5-3 or U, DBU and table 5-4 to find the actual FS values of In or Un.
Mean frequency of impulse output: fout = CPZ *<Reference Value> [Hz]

5.3 Operating mode (P4,P3,N4,N3,K4,K3,FFT,APM)

Cmd	Input format	Output format 1	Description
P4	**P4**	**P4=O**	**4-wire active power**

P3	**P3**	**P3=O**	**3-wire active power**

N4	**N4**	**N4=O**	**4-wire reactive power true wired (natural)**

N3	**N3**	**N3=O**	**3-wire reactive power true wired (natural)**

K4	**K4**	**K4=O**	**4-wire reactive power cross connected (artificial)**

K3	**K3**	**K3=O**	**3-wire reactive power cross connected (artificial)**

P4	P4=O	Set mode 4-wire active power
M	M=P4	Read actual mode, mode is 4-wire active power

FFT	**FFT<s>**	**FFT<s>=<O/E>**	**Fast Fourier Transformation x**

<s> Signal number for Fast Fourier Transformation (FFT)

<s>	**Signal**
1	U1
2	U2
3	U3
4	I1
5	I2
6	I3

Harmonics and wave form can be readout per phase with the commands RSI, RSU, RWI, RWU.
New results are available in the interval of the time base T.

FFT1	FFT1=O	Activate FFT of voltage U1
M	M=F1	Read actual mode, mode is F1 = FFT1

APM	**APM**	**APM=<0/1>**	**Read apparent power mode**
	APM<0/1>	**APM<0/1>=<O/E>**	**Set apparent power mode**

<0/1> 0: Mode 0: Calculation formula S = sqrt(P^2 + Q^2) is used
1: Mode 1: Calculation formula: S = Usum * Isum is used

APM	APM=0	Mode 0 active (default mode)
APM1	APM1=O	Set Mode 1

5.4 Error measurement (TF,TFX,TN,TNX,TU,TUX,CP,cp,CPX,cpX,FFS,FFSX,FF,FFX,STP)

Cmd	Input format	Output format 1	Description
TF	**TF**	**TF=<time>**	**Read error measurement time of input 1**
	TF<time>	**TF<time>=<O/E>**	**Set error measurement time of input 1**

TFX	**TFX<input>**	**TFX<input>=<time>**	**Read error measurement time**
	TFX<input>,<time>	**TFX<input>,<time>=<O/E>**	**Set error measurement time**

<time> Error measurement time in [s], string = 5 chars right aligned with leading blanks.
The error measurement is always running with impulses TU. Based on TF and the actual load conditions the number of impulses TU is defined to approximately reach the measuring time TF.
Note: Newly set <time> values become only valid after start command FF or FFX. If TU, TUX was activated before (default setting), the answer to a read command before execution of a start command will be =E.

<input> Selection of impulse input

<input>	Impulse input
0	All inputs (set command only)
1	Input 1

TF5	TF5=O	Set error measurement time TF = 5s.
FF	FF=O	Start error measurement, new settings overtaken
TF	TF= 5	Actual error measurement time of input 1 is 5s.
TFX1,20	TFX1,20=O	Set error measurement time of input 1 to 20s.
FFX1	FFX1=O	Start error measurement of input 1
TFX1	TFX2=…20	Actual error measurement time of input 1 is 20s

TN	**TN**	**TN=<N>**	**Read number of repetitions for input 1**
	TN<N>	**TN<N>=<O/E>**	**Set number of repetitions for input 1**

TNX	**TNX<input>**	**TNX<input>=<N>**	**Read number of repetitions**
	TNX<input>,<N>	**TNX<input>,<N>=<O/E>**	**Set number of repetitions**

<N> Number of repetitions of the same error measurement, range: 1 … 100.
The calculation of mean value and standard deviation is based on N measurements.
Note: Newly set <N> values become only valid after start command FF or FFX.

<input> Selection of impulse input

<input>	Impulse input
0	All inputs (set command only)
1	Input 1

TN5	TN5=O	Set number of repetitions of input 1 N = 5.
FF	FF=O	Start error measurement, new settings overtaken
TN	TN=5	Actual number of repetitions of input 1 is 5.

TU	**TU**	**TU=<imp>**	**Read number of test impulses of input 1**
	TU<imp>	**TU<imp>=<O/E>**	**Set number of test impulses of input 1**

TUX	**TU<input>**	**TU<input>=<imp>**	**Read number of test impulses**
	TU<input>,<imp>	**TU<input>,<imp>=<O/E>**	**Set number of test impulses**

<imp> Number of test impulses or revolutions of the error measurement,
Note: Newly set <imp> values become only valid after start command FF or FFX. If TF, TFX was activated before, the answer to a read command before execution of a start command will be =E.

<input> Selection of impulse input

<input>	Impulse input
0	All inputs (set command only)
1	Input 1

TU3	TU3=O	Set number of test impulses of input 1 TU = 3.
FF	FF=O	Start error measurement, new settings overtaken
TU	TU= 3	Actual number of test impulses of input 1 is 3.

CP	**CP**	**CP=<const>**	**Read constant [i/Ws] of impulse input 1**
	CP<const>	**CP<const>=<O/E>**	**Set constant [i/Ws] of impulse input 1**

Cmd	Input format	Output format 1	Description
CPX	**CPX<input>**	**CPX<input>=<const>**	**Read constant [i/Ws] of impulse input x**
	CPX<input>,<const>	**CPX<input>,<const>=<O/E>**	**Set constant [i/Ws] of impulse input x**

cp	**cp**	**cp=<const>**	**Read constant [i/kWh] of impulse input 1**
	cp<const>	**cp<const>=<O/E>**	**Set constant [i/kWh] of impulse input 1**

cpX	**cpX<input>**	**cpX<input>=<const>**	**Read constant [i/kWh] of impulse input x**
	cpX<input>,<const>	**cpX<input>,<const>=<O/E>**	**Set constant [i/kWh] of impulse input x**

<const> Meter constant of device under test of impulse input x with unit depending on command: CP, CPX [i/Ws], cp, cpx [i/kWh], answer to read commands: <const> = string with 8 chars.
Note: Newly set <const> values become only valid after start command FF or FFX.

<input> Selection of impulse input

<input>	Impulse input
0	All inputs (set command only)
1	Input 1

cpX1,5000	cpX1,5000=O	Set constant of input1 to 5000 i/kWh
FFX1	FFX1=O	Start error measurement of input 1
cpX1	cpX1=5000.000	Read actual constant of input 1, 5000 i/kWh
CPX1	CPX1=0.001389	Read actual constant of input 1, 0.001389 i/Ws

FFS	**FFS**	**FFS=<x>**	**Read ref. value for error meas. at input 1**
	FFS<x>[,<mode>]	**FFS<x>,<mode>=<O/E>**	**Set reference value for error meas. at input 1**

FFSX	**FFSX<input>**	**FFSX<input>=<x>**	**Read reference value for error measurement**
	FFSX<input>,<x>,<mode>	**FFSX<input>,<x>,<mode>=<O/E>**	**Set reference value for error measurement**

<x> Reference value for error measurement at input x, <x> can have the same values as at FOUT command, see table 5-5 in chapter 5.2.
Note: Newly set <x> values become only valid after start command FF or FFX.

<mode> Energy direction (+/-)
1 = The impulse output is independent of the energy direction
2 = The impulse output is only active when the reference values are positive (+)
3 = The impulse output is only active when the reference values are negative (-)

<input> Selection of impulse input

<input>	Impulse input
0	All inputs
1	Input 1

FFS	FFS=34	Read ref. value of input 1, reference = PΣ4
FFS38	FFS38=O	Set ref. value of input 1 to reference = QNΣ4
FFSX0	FFSX1=38	Read reference values of all impulse inputs reference input 1 = QNΣ4

FFM	**FFM**	**FFM=<mode>**	**Read the actual energy direction for all impulse inputs**
	FFM<mode>	**FFM<mode>=<O/E>**	**Set the energy direction for all impulse inputs**

FFMX	**FFMX<input>**	**FFMX<input>=<mode>**	**Read the actual energy direction**
	FFMX<input>,<mode>	**FFMX<input>,<mode>=<O/E>**	**Set the energy direction**

<mode> Energy direction (+/-)
1 = The impulse output is independent of the energy direction
2 = The impulse output is only active when the reference values are positive (+)
3 = The impulse output is only active when the reference values are negative (-)

<input> Selection of impulse input

<input>	Impulse input
0	All inputs
1	Input 1

Cmd	Input format	Output format 1	Description
FF	**FF**	**FF=<O/E>**	**Start / restart error measurement of input 1**
	FF0	**FF0=<O/E>**	**Stop error measurement of input 1**
	FF<ref>	**FF<ref>=<O/E>**	**Set reference for error meas. of input 1**

FFX	**FFX<input>**	**FFX<input>=<O/E>**	**Start / restart error measurement**
	FFX<input>,<ref>	**FFX<input>,<ref>=<O/E>**	**Set reference for error measurement**

<ref> Reference value for error measurement

<ref>	**Reference for error measurement**
1	P
2	Q
3	S
4	I^2*1 ohm (option)
5	U^2 / 1 Mohm (option)

<input> Selection of impulse input

<input>	**Impulse input**
0	All inputs
1	Input 1

Note: The start command FFor FFX<input> is only executed, if constant of <input> ≠ 0 (CP,CPX,cp,cpX ≠ 0) and test duration of <input> ≠ 0 (TU,TUX ≠ 0 or TF,TFX ≠ 0).

FFX1,1	FFX1,1=O	Set reference P for input 1
FFX0	FFX0=O	Start error measurement of all inputs

Example 1: Error measurement with input1
mode: active energy 4-wire, device under test (DUT) constant: IN1: 5000 imp/kWh (LED),; repetition of same measurement: IN1: 5, load: 3 x 240V, 5A, PF=1, 50Hz

Command sequence for direct control after power-on or reset of the module.

Input format	Output format 1	Description
MODE1	MODE1=O	Activate output format 1
DB1	DB1=O	Select 12A current inputs
I8,8,8	I8,8,8=O	Set current range 12A for all phases
U5,5,5	U5,5,5=O	Set voltage range 260V for all phases
MAN	MAN=O	Activate and fix ranges with manual range selection
cpX1,5000	cpX1,5000=O	Set constant of IN1: 5000 i/kWh
TFX1,15	TFX1,15=O	Set test duration of IN1: 15s
TNX1,5	TNX1,5=O	Set repetition rate at IN1: 5
FFSX1,34	FFSX1,34=O	Set ΣP as reference for input 1
FFX0	FFX0=O	Start error measurement of all inputs (preset values cpX, TUX, TNX, FFX are overtaken)

Example of results, when asking for new results with explicite command ?<No> (not all possible results shown)

?23	Ea,12.60511, . -	Read energy and error of IN1: Waiting for start impulse
?30	Eh,0.000000,--------, --------,--------,--------	Read statistical error results of input 1: Waiting for start impulse
?23	Ea,12.60511, --.--	Read energy and error of IN1: First measurement running
?30	Eh,1.000000,--------, --------,--------,--------	Read statistical error results of input 1: First measurement running
?23	Ea,6.003422,-0.05701	Read energy and error of IN1: Energy 1, Error 1
?30	Eh,2.000000,1.000000, -0.05701,0.000000,4.148804	Read statistical error results of input 1: Error 1, repetition 1 of 5, Em=E1, Es=0, f(IN1)
?23	Ea,5.996533,0.057817	Read energy and error of IN1: Energy 2, Error 2
?30	Eh,3.000000,2.000000, 0.000405,0.081192,4.153399	Read statistical error results of input 1: Error 2, repetition 2 of 5, Em, Es, f(IN1)
?30	Eh,4.000000,3.000000, -0.01774,0.065454,4.148230	Read statistical error results of input 1: Error 3, repetition 3 of 5, Em, Es, f(IN1)
?30	Eh,5.000000,4.000000, -0.02756,0.056935,4.148804	Read statistical error results of input 1: Error 4, repetition 4 of 5, Em, Es, f(IN1)
?23	Ea,6.003334,-0.05553	Read energy and error of IN1: Energy 5, Error 5
?30	Eh,6.000000,5.000000, -0.03315,0.050869,4.148804	Read statistical error results of input 1: Error 5, repetition 5, Em, Es, f(IN1)
?30	Eh,7.000000,5.000000, -0.03301,0.050783,4.148230	Read statistical error results of input 1: Error 6, repetition 5, Em, Es, f(IN1); Em, Es are calculated over the last N measurements

5.5 Energy registers (WREG,wreg,WRS)

Cmd	Input format	Output format 1	Description
WREG	**WREG<reg>**	**WREG<reg>=<W>**	**Read actual energy [Ws] of register <reg>**
	WREG<reg>,<W>	**WREG<reg>,<W>=<O/E>**	**Set energy [Ws] of register <reg>**

wreg	**wreg<reg>**	**wreg<reg>=<W>**	**Read actual energy [kWh] of register <reg>**
	wreg<reg>,<W>	**wreg<reg>,<W>=<O/E>**	**Set energy [kWh] of register <reg>**

<W> Energy value of register with unit depending on command: WREG [i/Ws], wreg [i/kWh]

<reg> Selection of register

<reg>	**Energy register**
0	All registers (set command only)
1	P+ (active positive)
2	P- (active negative)
3	Q+ (reactive positive)
4	Q- (reactive negative)
5	User register 1
6	User register 2
7	User register 3

WREG1	WREG1=2.39797e+07	Actual value of register P+: 2.39797e+07 [Ws]
wreg1	wreg1=6.66101933	Actual value of register P+: 6.66101933 [kWh]
wreg1,10000	wreg1,10000=O	Set register P+ initial value 10’000 [kWh]
wreg1	wreg1=10000.0010	Actual value of register P+: 10000.0010 [kWh]

WRS	**WRS<reg>**		**Read reference value of user register**
	WRS<reg>,<x>	**WRS<reg>,<x>=<O/E>**	**Set reference value for user register**

<x> Reference value for user register, <x> can have the same values as at FOUT command, see table 5-5 in chapter 5.2.

<reg> Selection of user register

<reg>	**Energy register**
0	All user registers (set command only)
5	User register 1
6	User register 2
7	User register 3

WRS5	WRS5=34	Reference of user register 5 is PΣ4.
WRS5,38	WRS5,38=O	Set reference of user register 5 to QΣ4.

5.6 Status information (VER,SG,CPZ,ROLI,ROLU,TIME)

Cmd	Input format	Output format 1	Description
VER	**VER**	**VER1=<name><s>#<SN><CR>** **VER2=<version><CR>** **VER3=<options><CR>** **VER4=<date>**	**Read version information**

<name> Type of instrument, string saved in parameter PAR0

<s> Space character

<SN> Serial number of the instrument, string saved in parameter PAR1

<version> Firmware version

<options> List of enabled options, string saved in parameter PAR2

<date> Date of firmware

VER	VER1=PTS2.3C #35099<CR> VER2=V1.18<CR> VER3=N<CR> VER4=07.04.2010<CR>	PTS2.3C #35099, unit with serial number V1.18, firmware version N, Net synchronisation available 07.04.2010, date of firmware

VER0	**VER0**	**PTSx.3C <version><date of firmware><time of firmware><CR>** **DSPx.3C <version><date of firmware><time of firmware><CR>** **EMFILE <version><CR>** **CFCARD <date of manufacture> <space>#<SN> <CR>**	**Read additional version information**
	VER0	PTSx.3C V1.18 B0001 compiled Apr 07 2010 09:28:18<CR> DSPx.3C V1.18 B0001 compiled Apr 7 2010 08:27:21<CR> EMFILE V2.30.20<CR> CFCARD 20070709 CF 256MB #TSS21953071022125350<CR>	Main firmware name, Firmware version, date and time of the Main-firmware DSP firmware name, Firmware version, date and time of the DSP-firmware File system, Filesystemversion CF Card, date of manufacture, max. space, CF Card serial number

SG	**SG**	**I=<i1><i2><i3><CR>** **U=<u1><u2><u3><CR>** **M=<mode><CR>** **CPZ=<const Fout1><CR>** **B=<AUT / MAN><CR>** **T=<time base><CR>** **TF=<error time><CR>** **G=0<CR>** **DB=<dbi><CR>** **DBU=<dbu>**	**Read status of instrument**

<ix>, <ux> Actual current and voltage ranges of phases x, x = 1,2,3. See command descriptions of I, U.

<mode> Actual operating mode: P4, P3, N4, N3, K4, K3, FFT1 … FTT6 = F1 … F6

<const Fout1> Constant of impulse output 1 [i/WS]

<AUT / MAN> Range status automatic (AUT) or manual (MAN) range selection active

<time base> Time interval for load value measurement

<error time> Time interval for error measurement

G=0 DC measurement disabled

<dbi>, <dbu> Selected current path, voltage path, see command description of DB, DBU

SG	I=111 U=333 M=P4 CPZ=6.25 B=AUT T= 5 TF=-???- G=0 DB=1 DBU=0	Current range for all phases 4mA (12A input) Voltage range for all phases 65V (standard input) Operating mode active 4-wire Reference meter constant 6.25 i/Ws Range selection = automatic Time base 5 s Time for error measurement not yet defined DC measurement disabled Current input = 12A Voltage input = standard input

Cmd	Input format	Output format 1	Description
CPZ	**CPZ**	**CPZ=<const>**	**Read actual constant of impulse output 1**
	CPZ<out>	**CPZ<out>=<const>**	**Read constant of impulse output <out>**

<const> Reference meter constant of impulse output in [Ws], string = 8 characters

<out> Impulse output number

<out>	**Impulse generator**
1	Impulse output 1

CPZ	CPZ=6.250000	Constant of impulse output 1: 6.25 i/Ws
CPZ1	CPZ1=6.250000	Same result as line above, output 1: 6.25 i/Ws

ROLI	**ROLI**	**ROLI=<FL1><FL2><FL3>**	**Read current phases overflow status**

ROLU	**ROLU**	**ROLU=<FL1><FL2><FL3>**	**Read voltage phases overflow status**

If a high priority result EQ (range overflow) has occurred, these commands can be used to localize the voltage and / or current phase, where the overflow has been detected.

<FLx> Flag of phase x, x = 1,2,3
0: normal, no overflow
1: overflow in phase Lx

?	EQ	High priority result range overflow
ROLI	ROLI=000	No overflow
ROLU	ROLU=001	Range overflow in phase 3

TIME	**TIME**	**TIME=DD.MM.YY<s> hh:mm:ss**	**Read date and time**
	TIME<DD>,<MM>,<YY>,<hh>, <mm>,<ss>	**TIME<DD>,<MM>,<YY>,<hh>, <mm>,<ss>=<O/E>**	**Set date and time**

Date and time of internal Real Time Clock (RTC)

DD Day
MM Month
YY Year
hh Hour (format 24h)
mm Minutes
ss Seconds
<s> Space character

TIME	TIME=22.03.00 13:16:48	Read actual date and time
TIME22,3,0,14,0,0	TIME22,3,0,14,0,0=O	Set new date and time: 22.03.00 14:00:00
TIME	TIME=22.03.00 14:00:03	Read actual date and time

5.7 Test (TBT,FT)

Cmd	Input format	Output format 1	Description
TBT	**TBT1**	**TBT1=<O/E>**	**Start time base test**
	TBT0	**TBT=##.###**	**Stop time base test, measured time**

##.### Time between TBT1 and TBT0 in s.

TBT1	TBT1=O	Start time base test
TBT0	TBT0=06.677	Stop time base test, measured time: 6.677s

FT	**FT<freq>**	**FT<freq>=<O/E>**	**Start output of <freq> at impulse output 1**
	FT	**FT=<O/E>**	**Stop output of <freq> at impulse output 1**

<freq> Fix frequency [Hz], fmax = 58500 Hz

With FT command the setting are changed back to previous settings of impulse output 1, defined by FOUT or F command.

?36	En,18675.60,18675.60,10780.2	Read actual frequency at impulse outputs
FT10000	FT10000=O	Set fix frequency at impulse output 1 = 10 kHz
?36	En,10000.00,18676.28,10779.82	Read actual frequency at impulse outputs
FT	FT=O	Stop fix freq. output, activate settings before
?36	En,18675.53,18675.53,10779.68	Read actual frequency at impulse outputs

5.8 Parametrization (PAC,PAM,PAR,TPM)

Cmd	Input format	Output format 1	Description
PAC	**PAC**	**PAC=<N>**	**Read parametrization counter**

<N> Number of activations of parametrization status PAM=1

PAC	PAC=1664	Parametrization activation counter = 1664

PAM	**PAM**	**PAM=<0/1>**	**Read status of parametrization mode**
	PAM<PW>	**PAM=<1/0>**	**Enable parametrization mode**
	PAM0	**PAM0=O**	**Disable parametrization mode, save parameters**

<0/1> Parametrization mode
0: Parametrization mode disabled
1: Parametrization mode enabled, protected parameters can be changed

<PW> Password: Correct entry of the password (restricted to MTE service personnel) enables the parametrization of protected parameters. The status PAM=1 remains until command PAM0 or next reset of the system.

PAR	**PAR<Nr>**	**PAR<Nr>=<value>**	**Read parameter**
	PAR<Nr>,<value>	**PAR<Nr>,<value>=<O/E>**	**Set or overwrite parameter**

<Nr.> Number of parameter

<value> Content of parameter, format: string
Protected parameters can only be changed, if parametrization mode is enabled PAM=1.

	PAR0	PAR0=PTS2.3C	Name of the instrument
	PAR1	PAR1=27592	Serial number of the instrument
TPM	**TPM1**	**TPM1 = <O/E>**	**Activate the transparent mode**

With this command the transparent mode will be activated and the readout the parameters of the internal reference standard is enabled.

Example: Read out reference standard parameters

TPM1 ; Activate the transparent mode
TPM1 = O
VER ; Read out the firmware version of the internal reference standard
SWS2.3 #35099<CR>
V1.05<CR>
-<CR>
10.06.2009<CR>
PAR4 ; Read out parameter number 4
120<CR>
TPM0 ; Stop the transparent mode

5.9 Communication mode (MODE)

Cmd	Input format	Output format 1	Description
MODE	**MODE**	**MODE=<0/1>**	**Read status of communication mode**
	MODE<0/1>	**MODE<0/1>=<O/E>**	**Set communication mode**

<0/1> Communication mode
0: Compatibility mode to older EMH equipment,
Output format: Commands are not repeated, no error message is returned in case of wrong commands
1: Extended mode
Output format: Commands are repeated in the form **<command>=<answer>**

<answer>	**Description**
<value>	Returned value, readout with command (e.g. PAR)
O	OK, return to successfully executed set command
E	Error, return to not executed set command
?	Command unknown

MODE0		Activate mode 0 (output format 0 (OF0))
U	233	Read voltage ranges (OF0)
PAR6	2	Read PAR6 (OF0)
P4		Set mode P4, no return (OF0)
P5		Invalid command, not executed, no return (OF0)
MODE1	MODE1=O	Set mode 1, O: OK, successfully done
U	U=232	Read voltage ranges
PAR6	PAR6=2	Read PAR6, command repeated
P4	P4=O	Set mode P4, O: OK, successfully done
P5	P5=E	Invalid command, not executed, E: Error returned
XXX	XXX=?	XXX is an unknown command

5.10 Other commands (PCD, BR)

Cmd	Input format	Output format 1	Description
PCD	**PCD<device>**	**PCD=<0/1>**	**Set standard device**

With this command we can define the receiver of ambiguous commands

PCD1	PCD1=O	Ambiguous commands goes to the reference standard
PCD2	PCD2=O	Ambiguous commands goes to the source

BR	**BR**	**BR=<baud rate>**	**Read baud rate**
	BR<baud rate>	**BR<baud rate>=<O/E>**	**Set baud rate**

Default settings after power-on: 19000 baud, 8 bit, no parity, 1 stop bit

<baud rate> Available values: 300, 600, 1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200 baud

BR	BR=19200	Actual baud rate is 19200 baud
BR9600	<24><134><152>f<24><30><NUL> <24><152><254><152><152><230><128>	Baud rate is changed in unit, but PC program is still at 19200 baud, therefore the outputformat is not valid -> change baud rate at PC to 9600 baud
BR	BR=9600	Actual baud rate is 9600 baud

5.11 Commands to enable, disable the PTS color system key pad (EK,DK)

Cmd	Input format	Output format 1	Description
EK	**EK**	**EK=<O/E>**	**Enable key pad**

DK	**DK**	**DK=<O/E>**	**Disable key pad**

The PTS color system keyd is enabled after power-on.
With the **DK** command the following PTS color system keys are locked:

- Push button (turning function of THB remains unlocked)
- Menu card keys
- Cursor keys
- Keypad

Manual operation of the PTS color system is blocked.

With the **EK** command or with a reset command **R** or power off/on the keys are enabled again.

The PTS color system acknowleges any status change of the keys with a double beep.

DK	DK=O	Disable keys, PTS answers with double beep, manual operation is locked
EK	EK=O	Enable keys, PTS answers with double beep, all keys can be used again

6. Command description for control of the internal power source

6.1 Readout of results

6.1.1 Types of results

There are two types of result classes to be distinguished:

- **Results within the result list:** The values can be readout explicitly with the command **?<No>**. The automatic check for new values in the result list can be done with the command **SP**. The individual results in the result list can be enabled or disabled for readout. This is done with the command **SP<No>**
- **Results with high priority:** These results cannot be readout explicitly. They are automatically generated by the system and they are treated with preference at the next readout of results.

Table 6-1 Result list

If not especially mentioned, the values in the result strings **<value>** have 8 characters filled up with spaces if necessary. If a value is below the starting value or out of range, 8 lines are displayed
'--------'. The result string starts always with an **E**. The second character represents the identification **<id>** of the result. All result strings are terminated with a carriage return **<CR>**.

Result ?<No>	Flag SP<No>	Results	Result-String E<id>[,<value>[,<value>…]]
1	1	**Phase currents** I1, I2, I3 [A]	E@,<I1>,<I2>,<I3>
2	2	**Phase to neutral voltages** U1 U2, U3 [V]	EA,<U1>,<U2>,<U3>
9	9	**Phase angle current to voltage** ϕui1, ϕui2, ϕui3 [°], ps ϕui1: phase angle I1 to U1 ps: phase sequence	EH,< ϕui1>,<ϕui2>,<ϕui3>,<ps> Phase sequence <ps>: <L>: Left <R>: Right <?>: Unknown
12	12	**Absolute angles of current** i1, i2, i3 [%]	EK,<i1>,<i2>,<i3>
13	13	**Absolute angles of voltage** u1,u2,u3 [%]	EL,<u1>,<u2>,<u3>

6.1.2 Readout of results (SP,?,SU,RDR,RSU,RSI,PAR,i,u,p)

Cmd	Input format	Output format 1	Description
SP	**SP<No>**	**SP<No>=<flag>**	**Read <flag> status of result <No>**
	SP<No>,<flag>	**SP<No>,<flag>=<O/E>**	**Set <flag> of result <No>**

<No> Flag number of result regarding result list, see table 6-1.
0: All results (set command only)

<flag> 0: Disable result in result list
1: Enable result in result list
2: Enable result exclusively in result list (all other results are disabled, selected result is enabled)

SP0,0	SP0,0=O	Disable all results in the result list
SP0,1	SP0,1=O	Enable all results in the result list
SP10,2	SP10,2=O	Enable exclusively error E1 in result list

?	**?<No>**	**E<id>[,<value>[,<value>…]]**	**Read out of result <No>**

<No> Result number regarding result list, see table 6-1
Empty: Readout of next enabled, unread result in result list (a high priority result, e.g. ER comes first)

<id> Identification sign for new value, see Tab 6-1

SP1,2	SP1,2=O	Enable exclusively the current values
SP2,1	SP2,1=O	Additional the voltage values
?	<CR>	No new result
?	<CR>	No new result
?	ER	Time base interval finished, new results available

Cmd	Input format	Output format 1	Description
	?	E@,1.000223,1.000112,1.000221	New current values
	?	EA,230.0033, 229,9988, 230.0001	New voltage values
	?	<CR>	No new result
	?	<CR>	No new result
	…	…	…

SU	**SU<0/1>**	**SU<0/1>=<O/E>**	**Start / Stop automatic sending of results**

<0/1> 1: Start internal check for new results with simulation of **?** command (10 ms interval). New results, which are enabled in the result list, are send out as soon as they are available.
0: Stop automatic sending of results.

Note: This command is forseen for test purposes with an ASCII terminal program (e.g. windows hyper terminal). It is not recommended, to use this command for controlled readout of a results in a program, because the allocation between a request and a received result is not clear.

RDR	**RDR**	**RDR= E<id>[,<value>[,<value>…]]**	**All momentary values and events read out at the same time**

The readings of theRDR command are simmular than the results of the ? commands.
(see chapter 6.1.2)

RSI	**RSI<ph>**	**RSI<Phase>=<NewFlag>,<Factor>[,<Spectrum>]**	**The spectrum of the actual switched on current values**

MODE1: RSI<Phase>=<NewFlag>,<Factor>[,<Spectrum>]
MODE0: RSI=<NewFlag>,<Factor>[,<Spectrum>]

<NewFlag> = '1' if there is a new spectrum or
'0' if the spectrum was readout already.

<Factor> = A scale factor, with which the following spectrum must be multiplied to get the original values.
<Spectrum> = A hex dump of the actual spectrum in the form:
<Real0><Imag0><Real1><Imag1><Real2>....<Real31><Imag31>

Each value is returned as 4-digit hex number in two's complement format. The original spectrum is calculated by dividing each value by 32767 and multiplying with <Factor>.

Example (25A, 10% of 3. harmonic are switched on):

RSI=1,25,0000003D7FFF0000000000000CCC0000000000000....<CR>
This contains the following spectrum:
<Factor> = 25

Real0 =	0x0000 * 25 / 32767 = 0		
Imag0 =	0x003D * 25 / 32767 = 0.04645	OW0 =	0.4647A
Real1 =	0x7FFF * 25 / 32767 = 25		
Imag1=	0x0000 * 25 / 32767 = 0	OW1 =	25A
Real2 =	0x0000 * 25 / 32767 = 0		
Imag2 =	0x0000 * 25 / 32767 = 0	OW2 =	0 A
Real3 =	0x0CCC * 25 / 32767 = 2.4999		
Imag3 =	0x0000 * 25 / 32767 = 0	OW3 =	2.4999 A (= ca. 10% OW1)

If the corresponding amplifier is not switched on, then the command returns no spectrum.
In this case the return is:
RSI=1,0<CR>

RSU	**RSU<ph>**	**RSU<Phase>=<NewFlag>,<Factor>[,<Spectrum>]**	**The spectrum of the actual switched on voltage values**

As RSI, but the spectrum of the voltage is returned

PAR	**PAR<number>**	**PAR<number>=<value>**	**Read out parameters**

Sets the parameter <Number> to the value <Value>. If <Value> is omitted, the actual value of parameter <Number> will be returned.

The parameters are divided in different classes:

1..99 : System parameters(general)

100..199 : Write all DSP parameters at once

200..299 : Parameters of the 1. DSP channel (range parameters, DSP specific PARs)

300..399 : Parameters of the 2. DSP channel ...

Cmd	Input format	Output format 1	Description
i	**i**	**E@,<I1>,<I2>,<I3>**	**Read out the actually current values**
	The results of this command are identical to the command ?1		
u	**u**	**EA,<U1>,<U2>,<U3>**	**Read out the actually voltage values**
	The results of this command are identical to the command ?2		
p	**p**	**EH,<Phi1>,<Phi2>,<Phi3>**	**Read out the angles between current and voltage**
	The results of this command are identical to the command ?9		

6.2 General settings (RAMPI,RAMPU,RAMP,T)

Cmd	Input format	Output format 1	Description
RAMPI	**RAMPI<Phase>**	**RAMPI[<Phase>[,<Ramp Duration>]]**	**Read out the current ramp duration for the next switch on and switch off of the source**
	RAMPI,[<Phase>[,<Ramp Duration>]]	**RAMPI[<Phase>[,<Ramp Duration>]]=<O/E>**	**Set the current ramp duration for the next switch on and switch off of the source**
	<Phase> = Definition fo the phase. Available are values from 1 to 3 for L1, L2, L3 <Ramp Duration> = Definition of the ramp time min.0.0 seconds to max. 600.0 seconds		
RAMPU	**RAMPU<Phase>**	**RAMPU<Phase>[,<Ramp Duration>]]**	**Read out the voltage ramp duration for the next switch on and switch off of the source**
	RAMPU,[<Phase>[,<Ramp Duration>]]	**RAMPU[<Phase>[,<Ramp Duration>]]=<O/E>**	**Set the voltage ramp duration for the next switch on and switch off of the source**
	<Phase> = Definition fo the phase. Available are values from 1 to 3 for L1, L2, L3 <Ramp Duration> = Definition of the ramp time min.0.0 seconds to max. 600.0 seconds		
T	**T**	**T,<Time>**	**The time base for the measurement of the actual values**
	T,<Time>	**T<Time>**	**Set the time base for the measurement of the actual values**
	<Time> = Definiton of the time base in seconds. (Min. 0.1 seconds – Max. 9999.9 seconds) **Note:** The time base is NOT influencing the regulation processes of the DSP's.		

6.3 Loadpoint commads (I,U,AI,AU,BI,BU,W,PH,FRQ,SET,OFF,ON,R)

Cmd	Input format	Output format 1	Description
I	**I[<Phase>[,<current>]]**	**I[<Phase>[,<current>]]=<O/E>**	**Set the current values for the source**

The set point value of the current of phase <Phase> will be set to the value <current>. If before an amplitude was set with AI, this value will be overwritten.

Without parameter <current>, the switched on current is indicated.

Without parameters, the switched on currents for all phases are indicated.

U	**U[<Phase>[,<voltage>]]**	**U[<Phase>[,<voltage>]]=<O/E>**	**Set the voltage values for the source**

The set point value of the voltage of phase <Phase> will be set to the value <voltage>. If before an amplitude was set with AU, this value will be overwritten.

Without parameter <voltage>, the switched on current is indicated.

Without parameters, the switched on currents for all phases are indicated.

AI	**AI[<Phase>[,<Amplitude>]]**	**AI[<Phase>[,<Amplitude>]]=<O/E>**	**Set the current of <Phase> to <Amplitude> in percentage of internal end of range value**

<Phase> = Definition fo the phase. Available are values from 1 to 3 for L1, L2, L3
<Amplitude> = Definition of the amplitude in % The amplitude must be in the range <PAR8>..<PAR9>.

If a set point value has been set before with U or I it will be overwritten.

If no <Amplitude> is given the switched on amplitude will be displayed.

If no <Amplitude> and no <Phase> is given the switched on amplitude for all three phases will be displayed.

AU	**AI[<Phase>[,<Amplitude>]]**	**AI[<Phase>[,<Amplitude>]]=<O/E>**	**Set the voltage of <Phase> to <Amplitude> in percentage of internal end of range value**

Note: The same as for command AI, differing the <Amplitude> of the voltage amplifier will be set/read..

BI	**BI[<Phase>[,<Range>]]**	**BI[<Phase>[,<Range>]]=<O/E>**	**Set the range <Range> as set point range for the current amplifier**

<Phase> = Definition fo the phase. Available are values from 1 to 3 for L1, L2, L3
<Range> = Definition of the new set point range

BI without the parameter <Range> the actual switched on current range of <Phase> will be displayed.

BI without parameter will display all actual switched on ranges.

BU	**BI[<Phase>[,<Range>]]**	**BI[<Phase>[,<Range>]]=<O/E>**	**Set the range <Range> as set point range for the voltage amplifier**

Note: The same as for command BI, differing only in the set point ranges and zero ranges of the voltage amplifier will be set/read.

W	**W[<Phase>[,<Angle>]]**	**W[<Phase>[,<Angle>]]=<O/E>**	**Set the phase angle between U<phase> and I<phase> (Phi) to <angle> in degree**

<Phase> = Definition fo the phase. Available are values from 1 to 3 for L1, L2, L3
<Angle> = Definition of the angle between U and I in degrees. The angle is limited to the range 0.0 .. <360.0 degree by modulus operation.

The command W without <Angle> the Phi value of the corresponding phase is returned.
W without parameter returns the phase angles between U and I of all phases

Cmd	Input format	Output format 1	Description
PH	**PH[<Phase>[,<Angle>]]**	**PH[<Phase>[,<Angle>]]=<O/E>**	**Set the absolute phase angle of voltage phase <Phase> to the value <Angle>.**

<Phase> = Definition fo the phase. Available are values from 1 to 3 for L1, L2, L3
<Angle> = Definition of the absolute phase angle between U and I in degrees. The angle is limited to the range 0.0 .. <360.0 degree by modulus operation.

The command PH without <Angle> the absolute phase angle of the corresponding voltage phase is returned.
PH without parameter returns the phase angles of all voltage phases.

FRQ	**FRQ[<Frequency>]**	**FRQ[<Frequency>]=<O/E>**	**Set the fundamental frequency of the generators**

If <Frequency> is in the range <PAR4> .. <PAR5>, then the preset frequency will be switched on with the next SET command.

If <Frequency> = 0 and the option line synchronization is available and enabled, the line synchronization mode will be activated with the next SET command. To use the function a line synchronized rectangular signal must be connected to the Sync-In of the master printed circuit board.

The command without parameter will return the actual switched on frequency or 0 if line synchronization mode is active.

SET	**SET**	**SET=1**	**Switch on the previously defined values**

At the startup following calculations will be realized:

Rms value	= set rms value or amplitude * set end of range value + amplitude(s) of harmonic(s)	(last U/I command) (last AU/AI command)
Range	= defined set point range or smallest possible range	(last BU/BI command) (no BU/BI yet or BU/BI<Ph>,0)
Frequency	= last set frequency or <PAR10>	(last FRQ command) (no FRQ command yet)
Angle	= Phase angle + Phi (only at current amplifiers) Curve form = Sinus + Harmonics	(last PH command) (last W command)

Special cases:

If the rms value is 0, the amplifier is switched off and the zero range is set.

If a packet steering was defined and no value was changed, then the packet steering is started.

If a ripple control was defined and no value was changed, then the ripple control is started.

OFF	**OFF**	**OFF=O**	**Switch off the generator**

OFF, OFF0: switch off the voltages and the currents with a ramp function.

OFF1: switch off immediately U and I amplifiers.

OFF2: switch off immediately only U amplifiers (for dosage with U)

OFF3: switch off immediately only I amplifiers (for dosage with I)

After switch-off the zero range is set.

ON	**ON**	**ON=O**	**Switch on the generator again with the old values after a OFF command.**

In contrast to the SET command, changed values (e.g. U, I, W, PH etc.) are not overtaken, the state before OFF will be restored.

ON1: switch on again U and I amplifiers.

ON2: switch on again only U amplifiers (for dosage with U).

ON3: switch on again only I amplifiers (for dosage with I).

R	**R**	**R=O**	**Reset the equipment**

Reset of the whole equipment. The software is rebooted and all not saved parameters are lost.

6.4 Waveform commands (OWI,OWU)

Cmd	Input format	Output format 1	Description
OWI	**OWI[<Phase>[,<Index>[,<Amp>,<Phi>]]]**	**OWI[<Phase>[,<Index>[,<Amp>,<Phi>]]]=O**	**Define the harmonics for the current amplifiers.**

<Phase> = Definition fo the phase. Available are values from 1 to 3 for L1, L2, L3
<Index> = Order number of harmonics.
0=DC part of the harmonic
1= The amplitude of the fundamental is set to 100% after each SET or OFF command
2…31= 2. to 31. harmonic (2x .. 31x the frequency of the fundamental)
<Amp> The amplitude of the harmonic in percent to the amplitude of the fundamental harmonic
<Phi> Definition of the phaseangle of the harmonic in degrees to the own zero crossing

Because the harmonics are added, the total rms value of the output current becomes higher.

If <Amp> and <Phi> are omitted, the setting of harmonic with order number <n> is returned.

If only <Phase> is sent with the command, all harmonics of this phase will be returned.

Without parameters all definitions of harmonics of all current phases are returned.

If for <Index> and <Amp> 0 is entered (OWI<Phase>,0,0), then all amplitudes of the harmonics 2 .. 31 are deleted. The next SET command will switch on the fundamental wave form only.

Amplitude and phase relation are evaluated depending on the system parameter 14,

PAR14=0 or PAR14=1: The harmonic definitions will be saved and started up with the next SET command

PAR14=2 or PAR14=3: The harmonic definitions will be start up directly

PAR14=0 or PAR14=2:
<Amp>: The desired harmonic amplitude in percentage of the amplitude of the fundamental,
e.g. 50Hz, fundamental = 230V, 10% of 3. Harmonic => 230V 50Hz + 23V 150Hz
It is also allowed to change the amplitude of the fundamental.
<Phi>: Phase displacement of the corresponding harmonic. The phase value <Phi> is related to the zero crossing of the corresponding harmonic and is for each harmonic in the range of 0 .. 360°.

PAR14=4: Now we can also change the amplitude of the fundamental. The whole spectrum will be switched on with the command SET in the way as it was defined, i.e. with a fundamental amplitude other than 100% the rms value of the generated wave form will be changed correspondingly.

Example (PAR14=4):

U1,50
OWU1,3,20,0
SET
Now 50Vrms fundamental + 20% of 3. Harmonics are switched on.

U1,50
OWU1,1,50,0
OWU1,3,20,0
SET
Now the fundamental amplitude is only 50%, i.e. 50% of 50V fundamental and 20% of 50V of 3. Harmonics are switched on.

PAR14=5: With this mode can also change the amplitude of the fundamental. The whole spectrum will be switched on with the command SET in the way as it was defined, admittedly the fundamental will be scaled again to 100% accordant all other harmonics including their DC parts.

Example (PAR14=5):

U1,50
OWU1,3,20,0
SET
Now 50Vrms fundamental + 20% of 3. Harmonics are switched on.

U1,50
OWU1,1,50,0
OWU1,3,20,0
SET
Because the fundamental amplitude isn't now 100%, the calculation will be factor 100/fundamental amplitude = 100/50 = 2. According this calculation the source starts up 50Vrms fundamental and **40% of 50V in 3. Harmonic**.

OWU	**OWU[<Phase>[,<Index>[,<Amp>,<Phi>]]]**	**OWU[<Phase>[,<Index>[,<Amp>,<Phi>]]]=O**	**Define the harmonics for the voltage amplifiers.**

Note: The same as for command OWI, differing in the harmonics for the voltage amplifiers are defined.

6.5 Status request (BSY,SE,STATI,STATU)

Cmd	Input format	Output format 1	Description
BSY	**BSY**	**BSY = <0/1>**	**Request the status of a SET command**

With the BSY command can be polled, whether a SET command has been completely executed and whether a new SET command will be accepted.

BSY=0: The SET command is accepted
BSY=1: The SET command is still executed

SE	**SE**	MODE1: **SE=ExxSxxxxx** (x = '0' or '1') MODE0: **ExxSxxxxx** (x = '0' or '1')	**Extended command to request the status of the amplifiers**

SE=E<Error><Warning><later additional flags possible>
S<DOS contact U><DOS contact I><Busy flag><Packet steering active>
<Ripple control active>< later additional flags possible >

<Error>	= Error flag, a DSP was switched off because of an error.
<Warning>	= Warning, a DSP is operated under overload conditions.
<DOS contact U>	= The dosage contact of voltage is open
<DOS contact I>	= The dosage contact of current is open
<Busy flag>	= A DSP is actually busy (e.g. after a SET command)
<Packet steering active>	= The packet steering is active at the moment
<Ripple control active>	= The ripple control is active at the moment

The single flags are evaluated as logical OR over all the available CPU printed circuit boards in the system. A set error flag means that at least one printed circuit board in the system is in the error status.

STATI	**STATI[<Phase>]**	**STATI[<Phase>] = <0…4>**	**Status of the current amplifiers**

STATI<Phase>=0 The DSP is switched off.
STATI<Phase>=1 The DSP is switched on.
STATI<Phase>=2 There is actually an overload situation.
STATI<Phase>=3 The DSP was switched off caused by overload.
STATI<Phase>=4 The DSP was switched off by DIS.

The status 3 and 4 can only be reset by the command OFF or by new switch on.

STATU	**STATU[<Phase>]**	**STATU[<Phase>] = <0…4>**	**Status of the voltage amplifiers**

As STATI, but the status of the voltage amplifiers is readout

6.6 Ripple control (RCSI,RCSIP,RCSU,RCSUP)

Cmd	Input format	Output format 1	Description
RCSI	**RCSI<phase>**	**RCSI<phase> = <0 / 000.0 / 000.1…100.0>**	**Request the status of ripple control of the current amplifiers**
	RCSI<phase>, 0	**RCSI<phase>, 0 = <O/E>**	**Stops a running ripple control test run of the current of the indicated phase**

The following results hereby can be returned:

- `RCSI<Phase>=0` : The ripple control test run was not initialized or has finished.
- `RCSI<Phase>=000.0` : The ripple control test run was initialized and can be started.
- `RCSI<Phase>=000.1 .. 100.0` : The ripple control test run is running and xxx.x % of the telegram are processed up to now.

RCSI<Phase>,<Duration>, <Amplitude>,<Frequency>	**RCSI<Phase>,<Duration>, <Amplitude>,<Frequency> = <O/E>**	**Define a ripple control impulse or a ripple control impulse pause of the ripple control string of the current amplifiers.**

A maximum of 250 table points can be defined for each phase. This means, that a maximum of 250 consecutive RCSI commands are allowed for each phase.

<Phase> is the number of phase 1..3
<Duration> defines the duration of a ripple control impulse or impulse pause in milliseconds.
<Amplitude> defines the amplitude of a ripple control impulse in ampere. The value for <Amplitude> must be in the range between 0 and actual active amplitude of fundamental. This means, the ripple control amplitude must be smaller than the amplitude of the fundamental.
For the generation of an impulse pause the amplitude must be set to 0.
<Frequency> is the desired frequency of the ripple control impulse.

After definition of a complete ripple control telegram with several RCSI commands, the ripple control test run can be started with the SET command.

When the test run is finished, the defined telegram is deleted.

Example:

At U1 a voltage of 230V,50Hz shall be switched on.

The following ripple control telegram shall be defined and sent:

S010101

S = Start impulse (10V, 600ms, 166.7Hz)
1 = Impulse (10V, 200ms, 166.7Hz)
0= Pause (0V, 200ms)

Switch on voltage:

```
U1,230
FRQ50
SET
```

Define telegram:

```
RCSU1,600,10,166.7
RCSU1,200,0,166.7
RCSU1,200,10,166.7
RCSU1,200,0,166.7
RCSU1,200,10,166.7
RCSU1,200,0,166.7
RCSU1,200,10,166.7
```

Start telegram:

```
SET
```

Cmd	Input format	Output format 1	Description
RCSIP	**RCSIP<Ph>,<Time>,<Amplitude>,<Freq>**	**RCSIP<Ph>,<Time>,<Amplitude>,<Freq>**	**Define a ripple control impulse or a ripple control impulse pause of the ripple control string of the current amplifiers.**
	RCSIP<phase>, 0	**RCSIP<phase>, 0 = <O/E>**	**Stops a running ripple control test run of the current of the indicated phase**

This command is working exactly the same as the corresponding RSCI command, but the amplitude is not entered in ampere, but in percentage of the actual switched on current amplitude.

RCSU	**RCSU<phase>**	**RCSU<phase> = <0 / 000.0 / 000.1…100.0>**	**Request the status of ripple control of the voltage amplifiers**
	RCSU<phase>, 0	**RCSU<phase>, 0 = <O/E>**	**Stops a running ripple control test run of the voltage of the indicated phase**
	RCSU<Phase>,<Duration>, <Amplitude>,<Frequency>	**RCSU<Phase>,<Duration>, <Amplitude>,<Frequency> = <O/E>**	**Define a ripple control impulse or a ripple control impulse pause of the ripple control string of the voltage amplifiers.**

As RCSI, but with this commands ripple control of voltage amplifiers is steered.

RCSUP	**RCSUP<Ph>,<Time>,<Amplitude>,<Freq>**	**RCSUP<Ph>,<Time>,<Amplitude>,<Freq>**	**Define a ripple control impulse or a ripple control impulse pause of the ripple control string of the voltage amplifiers.**
	RCSUP<phase>, 0	**RCSUP<phase>, 0 = <O/E>**	**Stops a running ripple control test run of the voltage of the indicated phase**

As RCSIP, but with this command ripple control of voltage amplifiers is steered.

6.7 Other commands (SKLI,TRI,PCD)

Cmd	Input format	Output format 1	Description
SKLI	**SKLI<Phase>[,<Flag>]**	**SKLI<Phase>[,<Flag>] = <O/E>**	**The relay to connect the big and small current connectors at the PTS x.xC can be steered.**

<Flag> = 0 opens the relay. (Using of the low and high current outputs)

<Flag> = 1 closes the relay. (Using only the high current outputs)

<Phase> defines, from which phase the relay should be steered.

<Phase> = 0 steers the relays of all current phases.

If a load point is switched on in the current ranges 1 or 2, then the relay is automatically opened to protect the hardware.

Note: This command can only be used for PTS x.xC up to 120A

TRI	**TRI<Mode>**	**TRI<Mode> = <0/1>**	**Change between 4- and 3-wire mode**

<Mode> 0 = 4-wire mode active
1 = 3-wire mode active

PCD	**PCD<device>**	**PCD=<0/1>**	**Set standard device**

With this command we can define the receiver of ambiguous commands

PCD1	PCD1=O	Ambiguous commands goes to the reference standard
PCD2	PCD2=O	Ambiguous commands goes to the source